International Institute of Information Technology
Bangalore (IIITB)

Vehicle Type Classification

Param Pandey (MT2026065)
Nabodaya Mete (MT2026058)
Prakhar Srivastava (MT2026071)
Varun Gupta (MT2026031)

Machine Learning Project Report

15 October 2026

Contents

Executive Summary **iii**

1 Introduction **1**

2 Dataset Overview **1**
2.1 Target Variable 1
2.2 Input Features 1

3 Preprocessing Pipeline **2**
3.1 Duplicate and conflicting-label checks 2
3.2 Handling Missing Values 2
3.3 Encoding of Categorical Features 2
3.4 Feature Scaling 2
3.5 Construction of a ColumnTransformer 3

4 Exploratory Data Analysis **3**

5 Feature Engineering **3**
5.1 Log-Transformed Features 3
5.2 Ratio Features 3
5.3 Interaction and Polynomial Features 4
5.4 Feature Ablation 4

6 Train–Validation Split **4**

7 Supervised Models (Full Dataset) **5**
7.1 Syllabus-compliant model families 5
7.2 Finalist five-fold evaluation, three shuffles 5

8 Final Model Selection **6**

9 Threshold Optimization **7**

10 Submission Artifact Verification **7**

11 Experiments with 20% Dataset **8**

12 Unsupervised Learning **8**

13 Key Insights & Discussion **8**

14 Limitations **9**

15 Future Work **10**

16 Conclusion **10**

References **11**

Appendix A. Public leaderboard submission history **11**

Appendix B. Artifact and source traceability **12**

List of Figures

1 Final Kaggle submissions and public leaderboard scores iii
2 Class distribution after deduplication 3
3 Training-set Pearson correlation matrix 4
4 Out-of-fold confusion matrix, winning ensemble 9

List of Tables

1 Dataset partitions: rows and predictors 1
2 Input features grouped by category 2
3 Retrospective three-fold screening (seed 42) 5
4 Five-fold evaluation of the finalist models 6
5 Final five Kaggle submissions 6
6 Pairwise label disagreement on test data 7
7 Effect of training-set size (20% vs. 100%) 8
8 Public leaderboard submission history 11

Executive Summary

The project develops a three-class vehicle classifier from 20 numeric specifications, compares course-taught supervised methods, and chooses five Kaggle submissions for final private evaluation. The supplied training data contains 10,000 rows, of which 2,500 repeat an existing predictor vector with a consistent label, leaving 7,500 distinct training examples. The competition test contains 2,500 unlabeled vehicles. After deduplication, stratified cross-validation and numeric scaling, penalized multinomial logistic models outperformed most Gaussian and tree-based alternatives.

The strongest observed public submission scored 0.90500. It averaged predicted probabilities from two models: 19-feature L2-regularized multinomial Logistic Regression and 20-feature Elastic Net-regularized multinomial Logistic Regression. A diverse shortlist of five includes both component models, an engineered-feature model, and an LDA-assisted blend. Repeated five-fold cross-validation is used to temper reliance on the public leaderboard. The best mean validation score is not guaranteed to be the best private score.

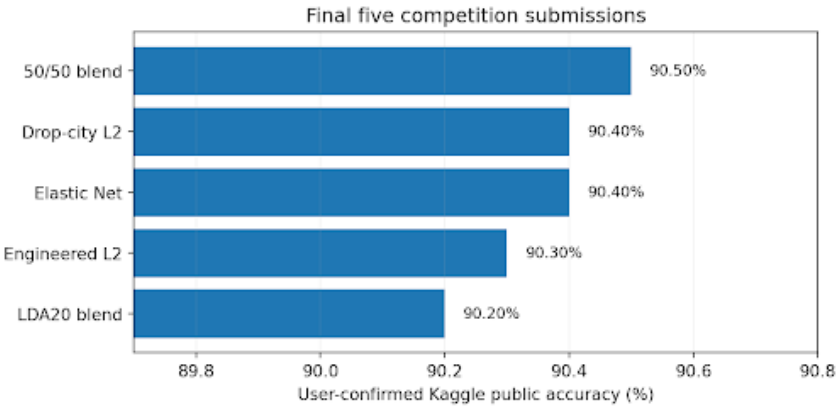


Figure 1: Five final Kaggle submissions and user-confirmed public leaderboard scores (not private scores).

Report at a Glance	
Project dataset	`vehicle_train_10000.csv` / `vehicle_test.csv`
Top observed public accuracy	90.50% (0.90500)
Finalist submissions	Five syllabus-compliant models and ensembles
Prepared	14 October 2026
Submission format	Report and reproducible Colab notebook

Scope statement: All reported Kaggle accuracies were supplied from completed public leaderboard submissions. No private-test labels or scores were available. Additional benchmarks performed for this report are explicitly identified as retrospective analyses.

1 Introduction

Vehicle classification from structured automotive specifications is a supervised multiclass learning problem. Each observation provides engine, powertrain, geometric, efficiency and vehicle-dynamics measurements; the objective is to estimate a class label in $\{0, 1, 2\}$. With 20 numeric inputs and a limited number of distinct labeled examples, the problem favors careful preprocessing, well-regularized models, leakage-resistant validation and interpretable error analysis.

The modeling program began with a Colab notebook comparing regularized least squares, Gaussian and discriminative classifiers, and Gaussian-noise augmentation. Follow-up experiments tested selected feature removal, Elastic Net, class-conditional methods, tree-based models, probability ensembles, threshold offsets and error-driven polynomial features. The final evaluation optimizes competition accuracy while respecting the course syllabus. The competition objective is to select five distinct submissions for evaluation on an unseen private test partition.

2 Dataset Overview

2.1 Target Variable

`vehicle_class` is an encoded categorical target taking values 0, 1 and 2. The dataset does not provide a verified semantic mapping such as sedan/SUV/sports vehicle, so this report uses only the numeric class codes. The `vehicle_id` column is a record identifier, never a predictor.

Dataset partition	**Rows**	**Predictors**	**Target available**
Training CSV (as supplied)	10,000	20	Yes
Unique labeled training vectors	7,500	20	Yes
Kaggle test CSV	2,500	20	No

Table 1: Dataset partitions used in the project, showing the number of rows, the number of predictors, and whether the target label is available.

The original training label distribution is 3,389 class-0, 3,326 class-1, and 3,285 class-2 rows; after deduplication, the class counts are 0: 2539, 1: 2490, 2: 2471. All classes are sufficiently represented for stratified splitting.

2.2 Input Features

All 20 predictors are numeric (floating point in the supplied CSVs). Physical names and units derive from the field names; no categorical one-hot encoding or text preprocessing was necessary.

Group	Dataset field names
Powertrain	`engine_displacement_l`, `horsepower_hp`, `engine_cylinders`, `torque_nm`, `battery_capacity_kwh`
Vehicle geometry	`vehicle_weight_kg`, `wheelbase_mm`, `vehicle_length_mm`, `vehicle_width_mm`, `vehicle_height_mm`
Usability / clearance	`ground_clearance_mm`, `turning_radius_m`, `cargo_capacity_l`, `tire_width_mm`
Energy and aerodynamics	`fuel_tank_capacity_l`, `city_fuel_efficiency_kmpl`, `highway_fuel_efficiency_kmpl`, `drag_coefficient`
Driving dynamics	`acceleration_0_100_kmh_s`, `braking_distance_m`

Table 2: The 20 numeric input features grouped by physical category, with their dataset field names.

3 Preprocessing Pipeline

3.1 Duplicate and conflicting-label checks

After excluding the identifier and label, 2,500 training records exactly duplicate another set of features. None of the duplicate feature vectors has conflicting labels, and no exact predictor-vector overlap was detected between the training and competition test files. We retain one copy per vector, avoiding duplicate leakage across CV folds and inadvertent weighting of repeated examples.

3.2 Handling Missing Values

There are zero missing predictor values in both supplied train and test files. Therefore, no imputation was applied in the historically submitted finalist models. A median-imputation stage is included in the final teaching pipeline as a robust option for future data, fitted inside the training partition rather than globally.

3.3 Encoding of Categorical Features

No predictor column is categorical. OneHotEncoder is therefore intentionally omitted in the executed pipeline. The target is already represented by numerical class codes, but it is modeled as categorical, not treated as a continuous response for final logistic classifiers.

3.4 Feature Scaling

Most finalist pipelines use StandardScaler to transform each numeric feature using a training-fold mean and standard deviation:

$$z_j = \frac{x_j - \mu_j}{\sigma_j}.$$

This gives penalized logistic models comparable feature scales. The scaler is refitted on each CV training fold, avoiding data leakage from held-out examples.

3.5 Construction of a ColumnTransformer

A demonstration ColumnTransformer contains a numeric branch `[SimpleImputer(strategy="median"), StandardScaler()]` and an empty categorical branch. Because the actual dataset is fully observed and numeric, the historical submissions use an equivalent StandardScaler-only pipeline for exact reproducibility. Tree models were evaluated without requiring standardization.

4 Exploratory Data Analysis

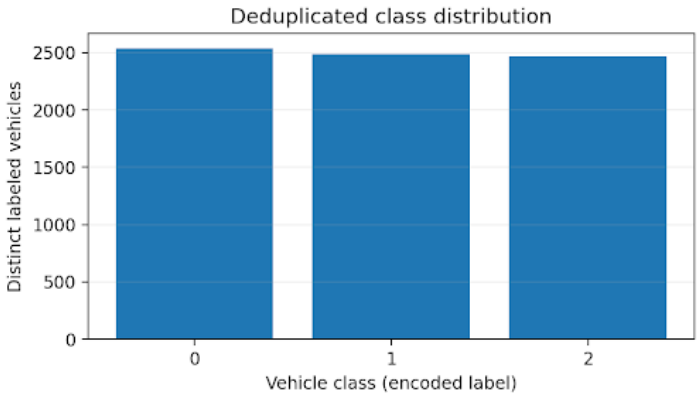


Figure 2: Balanced distribution across the three encoded classes after deduplication.

No severe target imbalance is evident. Most predictive information appears in specifications related to power, vehicle mass, dimensions, efficiency and tire configuration. A correlation review found redundant groups among physical specifications, encouraging regularization and cautious feature interaction engineering.

Exploratory plots use only the unique labeled training rows. Unlabeled test-set predictors were reviewed for gross distribution shifts but were not used to infer target labels. An adversarial train-vs-test classifier produced about 0.495 AUC in an earlier diagnostic, consistent with no readily detected shift under that test; it does not prove the populations are identical.

5 Feature Engineering

5.1 Log-Transformed Features

For syllabus completeness, the consolidated notebook demonstrates `log1p` transformations for nonnegative values such as horsepower, displacement and vehicle weight. The historical source notebooks do not document a separate, scored log-transform submission, so no claim of validated improvement or deterioration is made. Log-transformed features are not used in the selected finalists.

5.2 Ratio Features

The engineered-feature finalist adds exactly eight row-wise features to the 20 input columns: 1000× horsepower/weight; 1000× torque/weight; horsepower/displacement; city/highway fuel efficiency; wheelbase/vehicle length; height/width; $10^6\times$ weight/(length × width); and acceleration time × horsepower. These represent power density, body proportions, weight-per-footprint

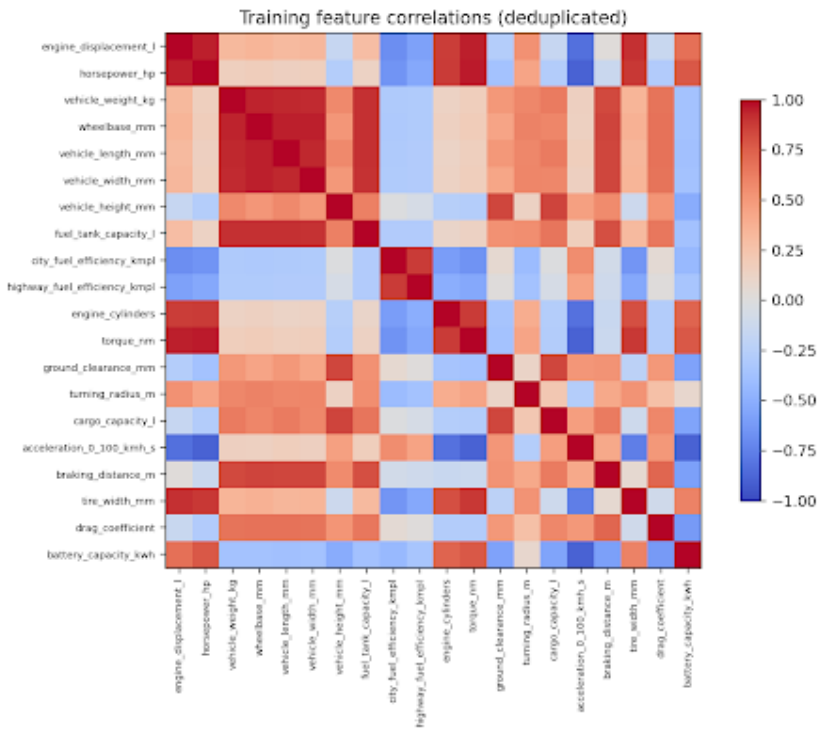


Figure 3: Training-set Pearson correlation matrix. Red and blue indicate positive and negative linear association, respectively.

and dynamic relationships. Each ratio is computed using only a row's observed inputs, so it does not leak labels or validation-fold statistics.

5.3 Interaction and Polynomial Features

Round 7 tested class-0-focused second-order interactions involving horsepower, vehicle weight, tire width, vehicle width and size, alongside a power-related feature block. A 1,875-row holdout showed an exploratory gain (89.55% vs 89.33%), but subsequent shuffled five-fold validation did not preserve it and the public submissions scored 90.20% and 90.10%. Cubic univariate features in Round 8 likewise offered no stable gain. The polynomial variants were therefore not retained in the top five.

5.4 Feature Ablation

Removing `city_fuel_efficiency_kmpl` improved the public score from 90.20% (all-feature L2 baseline) to 90.40% for the $C = 1$ model. Removing additional volume-related features did not generalize: the tested minus-volume model scored 90.00%. The winning ensemble pairs drop-city logistic with a full-feature Elastic Net model, balancing selective removal and regularized full information.

6 Train–Validation Split

Five-fold StratifiedKFold with shuffling and recorded random seeds is the main evaluation procedure. Folds are formed only after eliminating duplicate predictor vectors; fitted scalers and

model parameters are trained within the relevant fold. Out-of-fold class probabilities are then aggregated over all 7,500 unique records to compute classification accuracy, macro-F1, class-0 recall, log loss and confusion matrices.

Three repeated five-fold shuffles (seeds 42, 2026, 17) were recomputed for all five finalists using a consistent implementation. They cover the same set of labeled rows each time and are therefore *not* independent test datasets. Earlier experiments used more shuffles and a single 25% holdout to screen alternatives, but repeated examination of the same source records means the holdout is no longer strictly untouched across the overall project.

Accuracy = correct predictions / total cases; macro-F1 averages per-class precision–recall F1; log loss evaluates predicted probabilities. Public Kaggle leaderboard scores are reported separately and were not inserted into fold training. Private-test score and partition labels are not accessible.

7 Supervised Models (Full Dataset)

7.1 Syllabus-compliant model families

The original notebook compared Ridge/Lasso least-squares classifiers on one-hot encoded targets, Gaussian Naïve Bayes, LDA and regularized QDA, and L1/L2 multinomial Logistic Regression. Subsequent rounds included Elastic Net logistic, binary/hierarchical formulations, class-weight adjustments, decision trees, Random Forest, Extra Trees, quadratic/cubic features and ensembles. Extra Trees and Random Forest are included under tree-based classification methods; all final five use the explicitly permitted logistic/LDA methods and ensemble averaging.

Model (eligible screening)	**3-fold CV accuracy**	**3-fold macro-F1**
LDA shrinkage	89.39%	0.894
LDA	89.36%	0.894
L1 softmax (.1)	89.15%	0.892
QDA (reg .1)	88.96%	0.890
Random Forest	88.17%	0.882
Extra Trees	88.15%	0.882
Lasso least-squares	86.37%	0.862
Ridge least-squares	86.28%	0.861
Decision Tree	86.03%	0.862
GaussianNB	84.68%	0.848

Table 3: Retrospective three-fold screening, seed 42. These numbers are not directly comparable to the five-fold finalist metrics below. SVM/boosting methods from earlier broad screening were excluded as outside the listed syllabus.

7.2 Finalist five-fold evaluation, three shuffles

The LDA-assisted blend had the highest average accuracy among the finalists (89.369%), slightly above the simpler drop-city model (89.364%) and the public winner (89.351%). Such extremely small differences are not proof of generalization; they chiefly justify offering model-diverse submissions for private evaluation.

Model	Mean 5-fold acc.	Macro-F1	Class-0 recall	Log loss
Winner: 50/50 Logistic + Elastic	89.351%	0.8939	84.50%	0.26963
LDA-assisted 40/40/20 blend	89.369%	0.8940	84.31%	0.26913
Drop-city L2 Logistic	89.364%	0.8940	84.55%	0.27001
Full-feature Elastic Net	89.298%	0.8933	84.42%	0.26970
Engineered-feature L2 Logistic	89.329%	0.8937	84.67%	0.26986
Full-feature baseline L2	89.311%	0.8934	84.36%	0.27000

Table 4: Every model uses five-fold out-of-fold predictions on all 7,500 unique rows for seeds 42, 2026 and 17; the reported values average the three scores. They are not confidence intervals.

8 Final Model Selection

The five chosen entries balance observed public score, repeated cross-validation, reproducibility, and syllabus eligibility. To reduce private-test risk, the shortlist avoids filling all five slots with near-identical regularization-weight tweaks. It includes the two best individual models, one feature-engineered variant and an LDA-assisted blend alongside the public winner.

Priority	Exact final CSV	Public score	Mean repeated CV acc.
01	`top01_ensemble_905.csv`	0.90500	0.89351
02	`top02_dropcity_904.csv`	0.90400	0.89364
03	`top03_elastic_904.csv`	0.90400	0.89298
04	`top04_engineered_903.csv`	0.90300	0.89329
05	`top05_lda20_902.csv`	0.90200	0.89369

Table 5: The final five Kaggle submission files in priority order, with their public leaderboard scores and mean repeated cross-validation accuracy.

01. **50/50 drop-city L2 + full-feature Elastic Net** — Public winner; two independently fitted components.
02. **19-feature multinomial Logistic Regression** ($C = 1$) — Strongest simple model in repeated CV; no city fuel efficiency.
03. **20-feature Elastic Net Logistic Regression** ($C = 5$) — L1/L2 penalization; distinct regularization pattern.
04. **Logistic Regression + eight engineering features** — Distinct mechanical-ratio feature space.
05. **40/40/20 Logistic / Elastic Net / LDA blend** — Shared-covariance Gaussian discriminant component; best mean 3-seed CV.

Private-test selection remains uncertain. For example, the public winner and LDA-assisted blend differ in only 13 out of 2,500 test predictions, so the ensemble "diversity" is limited in absolute terms. No unobserved private-score claim is made. A sixth reserve, `reserve_baseline_902.csv` (L2 $C = 0.1$, 90.20% public), is available if the course requires exactly one distinct estimator per finalist rather than allowing multiple ensembles.

Pairwise changed labels	Winner	Drop-city	Elastic Net	Engineered	LDA20
Winner	0	18	11	17	13
Drop-city	18	0	29	33	23
Elastic Net	11	29	0	9	16
Engineered	17	33	9	0	16
LDA20	13	23	16	16	0

Table 6: Hard-label disagreement on the unlabeled 2,500-row competition test data; these do not indicate which model is correct.

9 Threshold Optimization

For multiclass probability outputs, class prediction normally uses argmax over class probabilities. Round 4 explored multiplicative per-class bias factors of the form $\arg\max_k [\, p_k(x) \times \exp(b_k) \,]$ and other rounds investigated class-0 sample weighting. Threshold bias could increase class-0 recall but often created additional false positives from classes 1 and 2. The public-tested class-bias candidate scored 90.00%, below the 90.50% unmodified ensemble.

A repeated-validation threshold scan improved mean accuracy by only about 0.03 percentage points at its best and was not robust to further evaluation. Because threshold search on many combinations can overfit validation data, the final five use the standard argmax decision rule with no additional score-calibrated class offsets.

10 Submission Artifact Verification

Each CSV contains exactly two columns, `vehicle_id` and `vehicle_class`, with 2,500 rows in the exact sample-submission ID order and labels restricted to $\{0, 1, 2\}$. The final notebook refits candidate models on all 7,500 distinct labeled vectors and creates the five named files shown in Section 8. It also writes a baseline reserve. The generation was checked against the original user-submitted CSVs and yielded **zero differing labels for all five selected files**.

The highest observed public score is 0.90500, obtained by `round4_blend_baseline_elastic.csv`. The selected files are identical in predictions to the historically submitted entries but are saved with `top01_...` through `top05_...` prefixes for easier administration. No Kaggle private evaluation has been performed. A full notebook execution was completed successfully in the provided environment.

11 Experiments with 20% Dataset

The original rounds did not establish a separate controlled 20%-training-data experiment. For this final report a supplemental study was performed and is not misrepresented as part of the initial work. From 7,500 unique examples, a stratified split allocates 6,000 rows to training and 1,500 to validation; a nested stratified 20% sample of the 6,000 training rows contains 1,200 examples. Both experiments use the same fixed 1,500-example validation partition.

Training partition	Train rows	Val. rows	Val. accuracy	Macro-F1
20% of training partition	1200	1500	88.07%	0.8814
100% of training partition	6000	1500	88.87%	0.8892

Table 7: Validation performance of the model trained on 20% versus 100% of the training partition, evaluated on the same fixed 1,500-example validation set.

The 1,200-example model scored 88.07% validation accuracy; the 6,000-example model scored 88.87%. The increase is consistent with benefit from more training records on this particular split, but it is not a learning-curve guarantee or a new, independent test of private generalization.

12 Unsupervised Learning

Principal component analysis (PCA) was considered as an unsupervised feature representation during exploratory model screening. PCA fits projections using feature covariance without target labels; a two-component projection appears in the final notebook for visualization. PCA-based supervised variants were not retained because they did not consistently improve accuracy. No independently verified clustering experiment or unsupervised competition submission exists in the supplied project files; the report does not invent K-means results.

This section therefore distinguishes unsupervised *exploratory dimensionality reduction* from final supervised classifiers. PCA is included solely to satisfy technical completeness and to clarify what was actually attempted.

13 Key Insights & Discussion

Most classification errors concern class 0. On the seed-42 winner OOF predictions, 392 actual class-0 rows were predicted as classes 1/2; another 380 class-1/2 rows were assigned class 0. Together, 772 of 792 total mistakes involve class 0. By comparison, class 1 versus class 2 mutual confusion contributed only 20 errors. A class-0-only recall objective is insufficient because improving true class-0 recovery can create more false positives in the other classes.

Regularized multinomial logistic regression handled this moderately sized structured dataset particularly well. Feature selection of city efficiency helped the simple model, while Elastic Net reduced redundancy using the full feature set. The winner combined these complementary descriptions and attained a 0.10-percentage-point public improvement over either component individually. However, tiny validation improvements did not consistently translate into public gains, and the observed public rankings differ slightly from mean repeated CV rankings.

The strongest technical lesson is evaluation discipline: remove duplicate feature rows, fit preprocessing inside folds, compare models on the same splits, distinguish probability log loss from accuracy, and limit decisions driven by repeated public-leaderboard feedback.

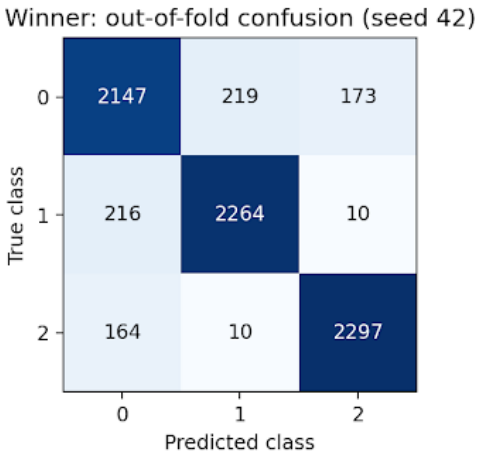


Figure 4: Out-of-fold confusion matrix for the 50/50 winning ensemble on five-fold CV seed 42 (7,500 distinct labeled rows).

14 Limitations

- The private-test labels and final private leaderboard results are unknown; any statement of private superiority would be speculative.
- The same 7,500 unique examples were reused across many historical rounds. Repeated model selection on these records can introduce meta-level overfitting despite fold-aware preprocessing.
- Several candidate models differ on only 9–33 of the 2,500 test rows; their apparent diversity is modest and may not protect against shift.
- The dataset provides encoded target labels without trusted semantic class descriptions, limiting physical interpretation of misclassified classes.
- Train and public-test distribution similarity tests do not prove there is no private distribution shift.
- Gaussian augmentation and polynomial engineering were tested but did not consistently improve clean accuracy.
- The original notebook contained a coding typo and unrelated Git/credential setup, both removed or corrected in the consolidated artifact.

15 Future Work

- Use an organizer-provided untouched final validation set or a preregistered nested CV plan before conducting further feature selection.
- Investigate additional valid measurements or data quality improvements that distinguish class 0 from overlapping classes 1 and 2.
- Assess probability calibration and model uncertainty under appropriate nested validation; do not optimize directly on public leaderboard outcomes.
- Analyze class-specific Gaussian feature subspaces and interpretable class-conditional covariance models under the course-approved methods.
- Conduct controlled learning curves over several training fractions and repeated splits, rather than relying on one 20% comparison.
- Revisit external test shift and class-prior sensitivity only when additional independent evidence becomes available.

16 Conclusion

This project consolidated the initial Colab notebook and eight subsequent rounds of systematic experimentation into one reproducible, syllabus-compliant supervised-learning workflow. The final dataset contained 7,500 distinct labeled vehicle examples and 2,500 competition test examples with 20 numeric predictors. We compared regularized least-squares classifiers, Gaussian Naïve Bayes, LDA/QDA, multinomial Logistic Regression, tree-based classifiers, handcrafted feature transformations, regularization variants and ensembles. Deduplication, stratified cross-validation, fold-aware scaling and class-wise error inspection were central to the modeling decisions.

The highest **observed public** Kaggle accuracy was 90.50%, produced by averaging 19-feature L2 multinomial logistic and 20-feature Elastic Net logistic probability estimates. Five top submissions were selected, spanning 90.20%–90.50% public accuracy, with a deliberate LDA-assisted alternative to improve coverage of modeling assumptions. The repeated validation results did not demonstrate a model that reliably exceeds 91%, and the private score cannot be inferred from public performance alone. The selected bundle is therefore intended to maximize transparent, defensible private-test selection rather than claim an unverified score.

References

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[2] R. Tibshirani, "Regression shrinkage and selection via the lasso," *Journal of the Royal Statistical Society: Series B*, vol. 58, no. 1, pp. 267–288, 1996.

[3] H. Zou and T. Hastie, "Regularization and variable selection via the elastic net," *Journal of the Royal Statistical Society: Series B*, vol. 67, no. 2, pp. 301–320, 2005.

[4] L. Breiman, "Random forests," *Machine Learning*, vol. 45, pp. 5–32, 2001.

[5] Course project source: Vehicle Type Classification competition data and original `Vehicle_Classification.ipynb`, with consolidated experiment notebooks and verified public scores supplied during project work, 2026. Kaggle competition: https://www.kaggle.com/competitions/vehicle-type-classification

Appendix A. Public leaderboard submission history

Phase	Historical submission filename	Public accuracy
Original augmented	`vehicle_submission_augmented.csv`	0.89600
Initial baseline	`vehicle_submission_baseline.csv`	0.90200
Initial blend	`vehicle_submission_blend.csv`	0.90200
Feature engineering	`vehicle_submission_engineered.csv`	0.90300
Drop-city L2	`vehicle_submission_drop_city.csv`	0.90400
Round 2 strong L2	`round2_drop_city_C005.csv`	0.89800
Round 2 LDA blend	`round2_lda_blend.csv`	0.90100
Round 2 regulation blend	`round2_regularization_blend.csv`	0.90000
Round 3 minus-volume	`round3_minus_volume_005.csv`	0.90000
Round 3 Elastic Net	`round3_elastic_full_5.csv`	0.90400
Round 4 50/50 blend	`round4_blend_baseline_elastic.csv`	0.90500
Round 4 class bias	`round4_bias_candidate.csv`	0.90000
Round 5 LDA20	`round5_lda20.csv`	0.90200
Round 5 reweight	`round5_base75_elastic25.csv`	0.90300
Round 6 tree10	`round6_tree_ensemble_10.csv`	0.90000
Round 6 tree20	`round6_tree_ensemble_20.csv`	0.90000
Round 7 focused poly	`round7_focused_poly_class0.csv`	0.90200
Round 7 power poly	`round7_power_poly_class0.csv`	0.90100

Table 8: Public leaderboard accuracy of every historical Kaggle submission, listed by experiment phase and submission filename.

Scores in this table are user-reported from completed Kaggle submissions. No private scores are claimed.

Appendix B. Artifact and source traceability

Project artifacts

Artifact	File	Notes
Main notebook	`Vehicle_Classification_Final_Course_Project.ipynb`	Reproducible workflow; includes the machine-learning portion of the original Colab notebook as a sanitized, historical appendix.
Selected CSV 01	`top01_ensemble_905.csv`	50/50 drop-city L2 + Elastic Net
Selected CSV 02	`top02_dropcity_904.csv`	Drop-city L2 Logistic
Selected CSV 03	`top03_elastic_904.csv`	Full-feature Elastic Net
Selected CSV 04	`top04_engineered_903.csv`	Engineered-feature L2 Logistic
Selected CSV 05	`top05_lda20_902.csv`	40/40/20 Logistic / Elastic Net / LDA blend
Reserve CSV	`reserve_baseline_902.csv`	Baseline L2 ($C = 0.1$)

Supporting files

- Original per-round experiment notebooks.
- A source-safe copy of the initial notebook.
- Historical validation CSVs.
- Newly computed three-seed CV, three-fold eligible-model screening and 20%-data comparison.

Reproduction validation

`Vehicle_Classification_Final_Course_Project.ipynb` executed end-to-end and each of the five generated finalist CSVs matched its source submission exactly in all 2,500 predicted labels and ID order. This is a reproducibility assertion about submitted predictions, not a private leaderboard measurement.